SE 226 – LAB#1 2025-2026 SPRING

Aim: Understanding basic input/output operations; the use of variables and arithmetic operators.

1. Create a **Python** project in which you have the following file: **myFirstLabScript.py**. This file will be your first Python script that asks the user's name and then greets the user. After that, the program will ask the user to enter his/her student id. Lastly, it will also print the id.

SAMPLE OUTPUT (bold parts are entered by user):

```
What is your name?
Ege
Hello Ege.
What is your Student ID?
20190120023
Your ID is 20190120023.
```

2. In a file named time_logic.py, ask the user to input a large integer representing a total number of seconds (e.g., 15000). In this task, you are expected to decompose a total number of seconds provided by the user, into hours, minutes, and seconds.

SAMPLE OUTPUT:

```
15000 seconds is 4 hours, 10 minutes, and 0 seconds.
```

3. In this task, you are required to develop a script named coordinate_geometry.py that calculates the straight-line distance between two points, (x_1,y_1) and (x_2, y_2) provided by the user. After capturing the four coordinate values as inputs, you must apply the Euclidean distance formula:

$$\boldsymbol{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

4. Try to output the below asterisk pattern using **the built-in function "print" only once**.

```
*******
 *****
  ***
   *
```

5. Do the previous tasks again, but this time in C++ programming language. You can use <cmath> library for mathematical operations.